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# 1 Intermediate Visualization in R

This chapter is based on a workshop that I ran at UC Davis in February 2019. We won't be covering it in class. But it gives some more background on visualization in R and presents some more advanced uses of ggplot.

To work through the plots, I recommend commenting out everything and adding one layer at a time. This is the best way to understand how each layer modifies the plot.

Some of the plots include alternative layers, mostly geoms, to illustrate how different ways of representing the data provide different visual affordances.

```
library(ggplot2)

data_dir = 'data'
data_files = c('papers' = 'paper_counts.Rds',
               'stations' = 'stations.Rds',
               'stations.utm' = 'stations.utm.Rds',
               'temperature' = 'temp.Rds')
data_paths = file.path(data_dir, data_files)
names(data_paths) = names(data_files)
data_urls = paste0('https://github.com/dsidavis/RGraphicsWorkshop/raw/master/data/', data_files)

## The repo includes a script `climate_data.R` that creates the final 3 datasets

## `if` isn't vectorized, so we have to loop over this manually
download_file = function(path, url) {
    if (!file.exists(path)) {
        download.file(url, path)
    }
}
purrr::walk2(data_paths, data_urls, download_file)
```

## 1.1 Base graphics vs. ggplot2

- R comes with a standard set of visualization tools, called *base graphics*.[1]

- Base graphics' approach to visualization is often called an "artist's palette" model
  - A set of low-level tools
    - `points()`, `lines()`, `polygon()`
    - `abline()`, `curve()`
  - Define a canvas first, then use these tools to paint elements on the canvas

[1] Strictly speaking, `base` is the name of a particular package; "base graphics" aren't included in this package, but instead are in the `graphics` package. So "base graphics" is strictly incorrect. But "graphics graphics" sounds silly, and the `graphics` package is loaded automatically when you launch R, so everyone just calls it "base graphics."

    - The common Python plotting packages (`matplotlib`, `seaborn`, etc.) are object-oriented variations on this
        * `matplotlib` was originally designed to work like plotting in MATLAB, in order to ease the transition for scientists coming from MATLAB
- Advantages of base graphics
    - One-liners
        * `plot(some_object)` often does something useful
            · Try it with dataframes, regression models, principal components results, hierarchical clusterings, igraph networks, …
        * `hist()`, `pairs()`, `barplot()`, `boxplot()`, `pie()`(!)
    - Straightforward to extend to custom classes
    - No* dependencies; works on any* R installation
    - Very flexible for unusual visualizations or combinations of plots
        * triplots/barycentric coordinates
        * scatterplots with marginal densities
        * heatmaps with marginal dendrograms
        * map with inset plots of local data
- Disadvantages of base graphics
    - Parameters aren't perspicuous
        * fill color is `bg` (background?)
        * point shape is `pch` (uhhh point-character?)
        * point and text size is `cex` (???)
    - "Ugly defaults"
    - Plots as side effects

- Advantages of `ggplot2`
    - Conceptual model
    - Facets
    - Themes and palettes
    - Plots are first-class objects
    - "Pretty defaults" (though I disagree)
- Disadvantages of `ggplot2`
    - Only* for data frames in *long format*
    - Programming with `ggplot2` requires understanding tidy evaluation

## 1.2 Plot 1: Dumbbell plot

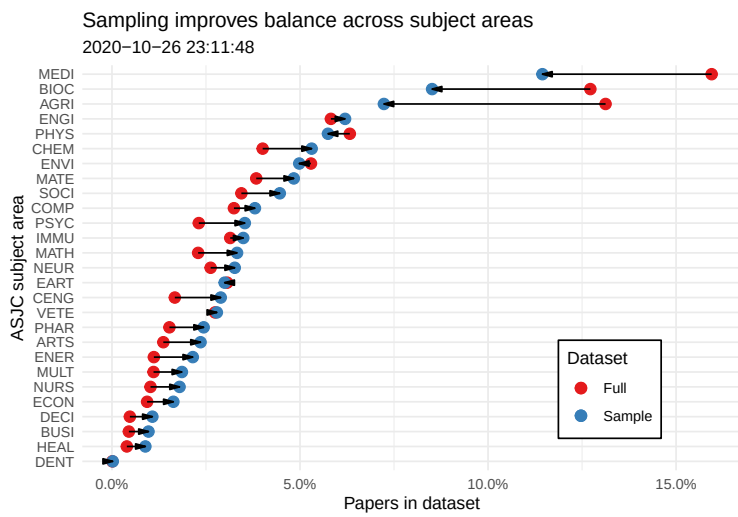
- @HicksImpactingCapabilitiesConceptual2018 figure 2

```
counts_df = readRDS(data_paths['papers'])

counts_df
```

```
## # A tibble: 27 x 3
##    subject_area   full sample
##    <fct>         <dbl>  <dbl>
##  1 MEDI         0.159  0.114
##  2 BIOC         0.127  0.0851
##  3 AGRI         0.131  0.0723
##  4 ENGI         0.0582 0.0620
##  5 PHYS         0.0632 0.0574
##  6 CHEM         0.0401 0.0531
##  7 ENVI         0.0529 0.0498
##  8 MATE         0.0384 0.0483
##  9 SOCI         0.0344 0.0446
```

```
## 10 COMP         0.0324 0.0380
## # ... with 17 more rows
```

```r
ggplot(counts_df, aes(subject_area)) +
    ## geoms
    geom_point(aes(y = full, color = 'Full'), size = 3) +
    geom_point(aes(y = sample, color = 'Sample'), size = 3) +
    geom_segment(aes(x = subject_area, xend = subject_area,
                     y = full, yend = sample),
                 arrow = arrow(angle = 15, type = 'closed',
                               length = unit(.025, 'snpc'))) +
    ## scales
    xlab('ASJC subject area') +
    scale_y_continuous(labels = scales::percent_format(),
                       name = 'Papers in dataset') +
    scale_color_brewer(palette = 'Set1',
                       name = 'Dataset') +
    coord_flip() +
    ## annotations and theme
    ggtitle('Sampling improves balance across subject areas',
            subtitle = Sys.time()) +
    # annotate(geom = 'label', x = 'HEAL', y = .12, label = Sys.time()) +
    theme_minimal() +
    theme(legend.position = c(.8, .2),
          legend.background = element_rect(fill = 'white'))
```




```r
# ggsave(paste0(plots_dir, 'plot_1.png'), height = 7, width = 7)
```

### 1.2.1 Discussion

1. What design decisions are being used to highlight the message of this plot?

2. What alternative designs could be used to communicate the same message?
   - Are any of these alternative designs better than the one I used?

   - Try implementing your idea in ggplot
3. How does this plot omit, distort, or mislead?

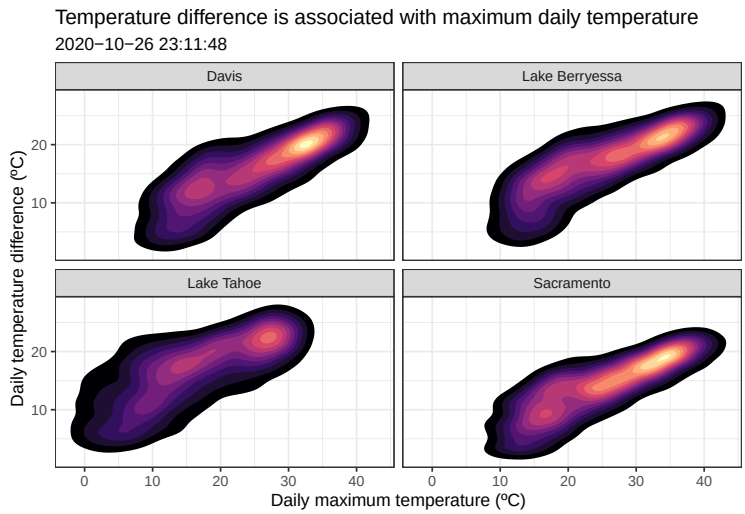
## 1.3 Plot 2: High-density scatterplots

```r
## Code to construct this dataset is in `climate_data.R`
temp_df = readRDS(data_paths['temperature'])

## 15k observations lead to lots of overplotting and long render times
str(temp_df)
```

```
## 'data.frame':    14577 obs. of  12 variables:
##  $ id           : chr  "USC00042294" "USC00042294" "USC00042294" "USC00042294" ...
##  $ name         : chr  "Davis" "Davis" "Davis" "Davis" ...
##  $ date         : Date, format: "2009-01-01" "2009-01-02" ...
##  $ year         : num  2009 2009 2009 2009 2009 ...
##  $ month        : chr  "January" "January" "January" "January" ...
##  $ day          : num  1 2 3 4 5 6 7 8 9 10 ...
##  $ temp_min     : num  3.3 3.9 0.6 -2.8 -2.2 1.7 0 3.9 0.6 -2.8 ...
##  $ temp_max     : num  6.1 6.7 10.6 11.1 10 7.2 12.8 7.2 10 16.7 ...
##  $ temp_delta   : num  2.8 2.8 10 13.9 12.2 5.5 12.8 3.3 9.4 19.5 ...
##  $ precipitation: num  0 0 0 0 5 8 0 0 0 0 ...
##  $ lat          : num  38.5 38.5 38.5 38.5 38.5 ...
##  $ lon          : num  -122 -122 -122 -122 -122 ...
```

```r
ggplot(temp_df, aes(temp_max, temp_delta)) +
    ## geoms
    # geom_point() +
    # geom_count() +
    # geom_bin2d() +
    # geom_hex(aes(color = after_stat(count))) +
    # geom_density2d(size = 1) +
    # stat_density2d(contour = FALSE, geom = 'raster',
    #                aes(fill = after_stat(density)),
    #                show.legend = FALSE) +
    stat_density2d(geom = 'polygon',
                   aes(fill = after_stat(level)),
                   show.legend = FALSE) +
    ## facets
    facet_wrap(~ name) +
    ## Annotations are repeated within each facet
    # annotate(geom = 'label', x = 200, y = 5, label = Sys.time()) +
    ## scales
    xlab('Daily maximum temperature (ºC)') +
    ylab('Daily temperature difference (ºC)') +
    scale_color_viridis_c(option = 'A') +
    scale_fill_viridis_c(option = 'A') +
    ## themes
    theme_bw() +
    ggtitle('Temperature difference is associated with maximum daily temperature',
            subtitle = Sys.time())
```

```
## Warning: Removed 104 rows containing non-finite values (stat_density2d).
```




```
# ggsave(paste0(plots_dir, 'plot_2.png'), height = 7, width = 7)
```

### 1.3.1 Discussion

1. What are the strengths and weaknesses of each different option?

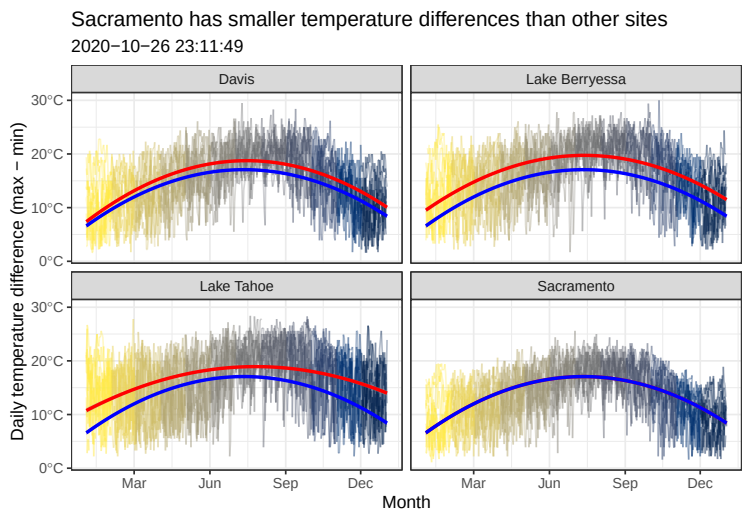
## 1.4 Plot 3: Time series + model predictions

```
## Hack to get month labels on day-as-year
## <https://stackoverflow.com/a/39861306/3187973>
temp_df$day.date = as.Date(temp_df$day, '2009-01-01')

## Model **for Sacramento alone**
model = lm(data = temp_df, temp_delta ~ day + I(day^2),
           subset = name == 'Sacramento')

## The director of the center where I gave this talk (who was also my postdoc supervisor) is part of th
## This is a good illustration of how tidyverse makes R more functional
predictions = predict(model, interval = 'confidence', level = .95)
predictions = as.data.frame(predictions)
predictions$day = model$model$day
predictions$day.date = as.Date(predictions$day, '2009-01-01')

ggplot(temp_df, aes(day.date, temp_delta)) +
    ## geoms
    geom_path(aes(color = month, group = year), alpha = .3, show.legend = FALSE) +
    ## Separate models for each panel
    geom_smooth(method = lm, formula = y ~ x + I(x^2),
                color = 'red', size = 1) +
    ## Sacramento-only model
    ## Use the smooth geom to get curve + ribbon
    ## But set `stat = 'identity'` because we don't want to fit a (new) model
    geom_smooth(data = predictions, stat = 'identity',
                aes(y = fit, ymax = upr, ymin = lwr),
                color = 'blue', fill = 'blue', size = 1) +
    ## facets
    facet_wrap(~ name) +
```

```r
    ## scales
    scale_x_date(name = 'Month', date_breaks = '3 month', date_labels = '%b') +
    scale_y_continuous(name = 'Daily temperature difference (max - min)',
                       labels = scales::math_format(expr = .x*degree*C)) +
    scale_color_viridis_d(limits = month.name,
                          option = 'E',
                          direction = -1) +
    # coord_polar() +
    ## themes
    theme_bw() +
    ggtitle('Sacramento has smaller temperature differences than other sites',
            subtitle = Sys.time())
```

```
## Warning: Removed 104 rows containing non-finite values (stat_smooth).
```

```
## Warning: Removed 1 row(s) containing missing values (geom_path).
```




```r
## Note effect of aspect ratios on perception
# ggsave(paste0(plots_dir, 'plot_3.png'), height = 7, width = 7)
# ggsave(paste0(plots_dir, 'plot_3a.png'), height = 7, width = 12)
```

### 1.4.1 Discussion

1. What are the dangers of using models internal to the plot?

2. Aspect ratios change our perception of the plot. Does this mean one aspect ratio is misleading?

## 1.5 Plot 4: Visualizing spatial data in R

- I don't recommend visualizing spatial data until you've learned a bit about coordinate systems and projections
- Definitely don't try to do math on spatial coordinates unless you know what you're doing!
- The `sf` package for working with spatial data has a ton of dependencies, so I'm going to disable all of the code blocks below.
  - They're here for you to explore on your own if interested.

  - **This code has not been checked for errors.**
- As of version 3, `ggplot2` can plot vector spatial data from `sf` objects using `geom_sf()`

- This section looks at two different ways of adding basemaps under `geom_sf()` plots

```
if (packageVersion('ggplot2') < package_version('3.0.0')) {
    stop('ggplot2 version 3 or later is required for this section')
} else {
    library(sf)
    library(ggmap)
    library(ggspatial)
}

stations = readRDS(data_files['stations'])
stations.utm = readRDS(data_files['stations.utm'])

stations
```

### 1.5.1 ggmap approach

- Services other than Google Maps are deprecated
- Requires Google account, API key
- Doesn't seem to play nicely with coordinates other than lat/lon
- See `?register_google` for links and instructions
- ***DO NOT SHARE YOUR API KEY***

```
# register_google(key = 'notarealkey', write = TRUE)
if (!has_google_key()) {
    stop('You need a Google Maps API key.  See ?register_google')
}

basemap = get_map(location = c(-121.4183, 38.5552),
                  maptype = 'terrain',
                  zoom = 8)

ggmap(basemap) +
    geom_sf(data = stations, inherit.aes = FALSE) +
    geom_sf_label(data = stations,
                  aes(label = name), nudge_y = -.1, size = 3)
# ggsave(paste0(plots_dir, 'plot_4_ggmap.png'), height = 7, width = 7)

## Example with UTM coordinates
# ggmap(basemap) +
#     geom_sf(data = stations.utm, inherit.aes = FALSE)
```

### 1.5.2 ggspatial approach

- `annotation_map_tile()` is slow, even with cached images
- See `rosm::osm.types()` for a list of basemap types

```
ggplot(data = stations) +
    annotation_map_tile(type = 'osm', zoom = 10) +
    geom_sf_label(data = stations,
                  aes(label = name), nudge_y = .03, size = 3) +
    geom_sf() +
    annotation_scale(location = 'tl') +
    annotation_north_arrow(location = 'br', which_north = 'true',
                           style = north_arrow_minimal)
# ggsave(paste0(plots_dir, 'plot_4_ggspatial.png'), height = 7, width = 7)
```

```r
## Example with UTM coordinates
# ggplot(data = stations.utm) +
#     annotation_map_tile(type = 'osm', zoom = 10,
#                         cachedir = system.file('rosm.cache', package = 'ggspatial')) +
#     geom_sf_label(data = stations,
#                   aes(label = name), nudge_y = 1000, size = 3) +
#     geom_sf()
```